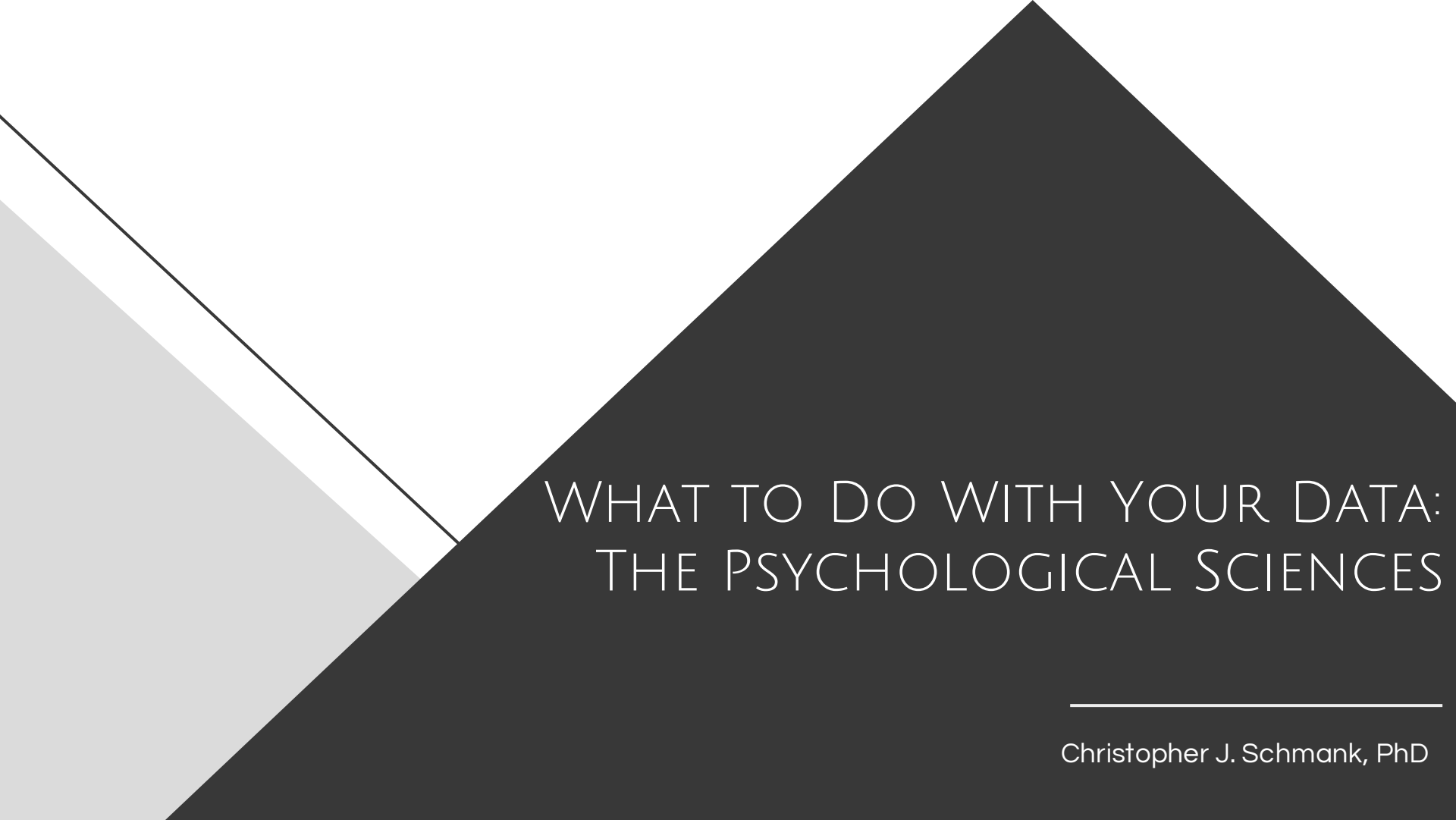




WHAT TO DO WITH YOUR DATA: THE PSYCHOLOGICAL SCIENCES

Christopher J. Schmank, PhD

OVERVIEW

- 01 BRIEF INTRODUCTION
- 02 CHALLENGES OF PSYCHOLOGICAL DATA
- 03 RESEARCH AS "STORYTELLING"
- 04 THE USUAL SUSPECTS (I.E., STATISTICS)



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CURRENTLY TEACHING

PSY 3120

Cognitive Psychology

PSY 4060/4840

Capstone Research
Methodology and Statistics

PSY 5090

Psychometric Theory

PSY 5790

Applied Univariate and
Bivariate Statistics

PSY 6500

Applied Multivariate and
Multivariable Statistics

REGULATION

How does regulating our emotions or stress impact our thinking and abilities?

RATIONALITY

What makes a rational thinker and how does this differ from human intelligence?

RESEARCH INTERESTS

TIP-OF-THE TONGUE STATES & PHANTOM RECOLLECTION

What influences our language production abilities or errors?

STATISTICAL MODELS OF COGNITIVE ABILITY

Psychometric network and latent variable modeling

CHALLENGES OF COLLECTING PSYCHOLOGICAL DATA

Measuring the mind and
unobservable phenomena

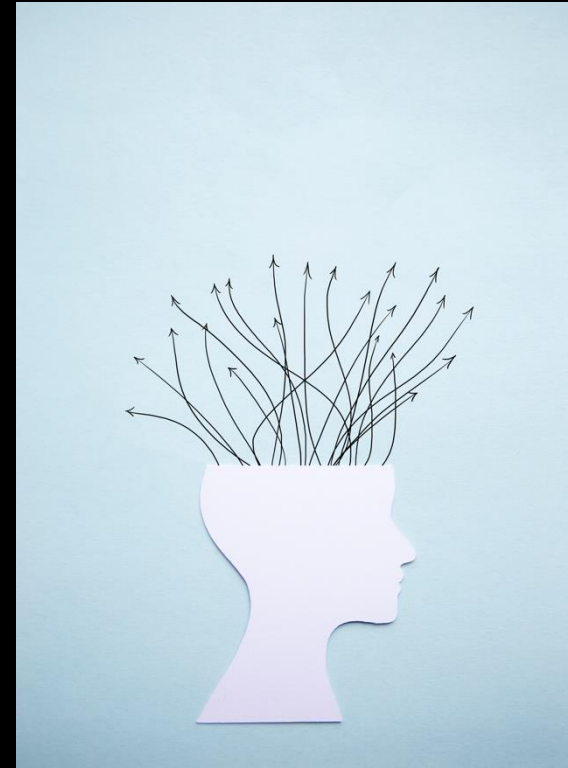


Data collection can involve:

- Attitudes
- Emotions
- Knowledge
- Memory
- Ability

Most variables are *latent* or
NOT directly observable!

How do we measure what
cannot be observed?



MANIFEST VARIABLES

- Measured variables
- Reaction Times

LATENT VARIABLES

- Implied variables
- Processing/Perceptual Speed
- How would you measure “Academic Burnout”?

RESEARCH AS “STORYTELLING”

Connecting our research question to our
scientific methodology and statistics

A COGENT "STORY"

RESEARCH
QUESTION



RESEARCH
METHODOLOGY



STATISTICAL
INFERENCE

Research questions ...

1. Start with a broad topic
2. Conduct literature review
3. Develop some “how” and “why” questions
4. Focus on the **FINER** things
Feasibility, Interesting, Novelty,
Ethical, Relevant
5. Establish hypotheses

THEN we can determine
appropriate research method



EXPERIMENTAL METHODS

1+ *manipulated* independent variable

High- vs. Low-Stress

200 mg vs. 95 mg vs. 0 mg Caffeine

QUASI-EXPERIMENTAL METHODS

1+ *manipulated* and 1+ *grouping*
independent variable

Regulation vs. No-Regulation

Younger- vs. Older-Adults

PSYCHOLOGICAL SCIENCE METHODOLOGIES

INDIVIDUAL DIFFERENCES (I.E., CORRELATIONAL)

No *manipulated* variables, but several
manifest (*predictor*) variables

Relationship Among Cognitive Abilities
and Academic Performance

Processing Speed, Reasoning, Memory

LONGITUDINAL METHODS

Research conducted on same
participants across time

Depression symptom change
across emerging adulthood

Day 1: Regulation

Day 2: No Regulation



HOW DOES STRESS IMPACT MEMORY?

Individual Differences Research:

- Collect stress inventory
- Collect memory performance

Experimental Research:

- Manipulate stress-levels
(e.g., Cold Water Pressor Task)
 - High- vs. Lower-Stress
- Collect memory performance



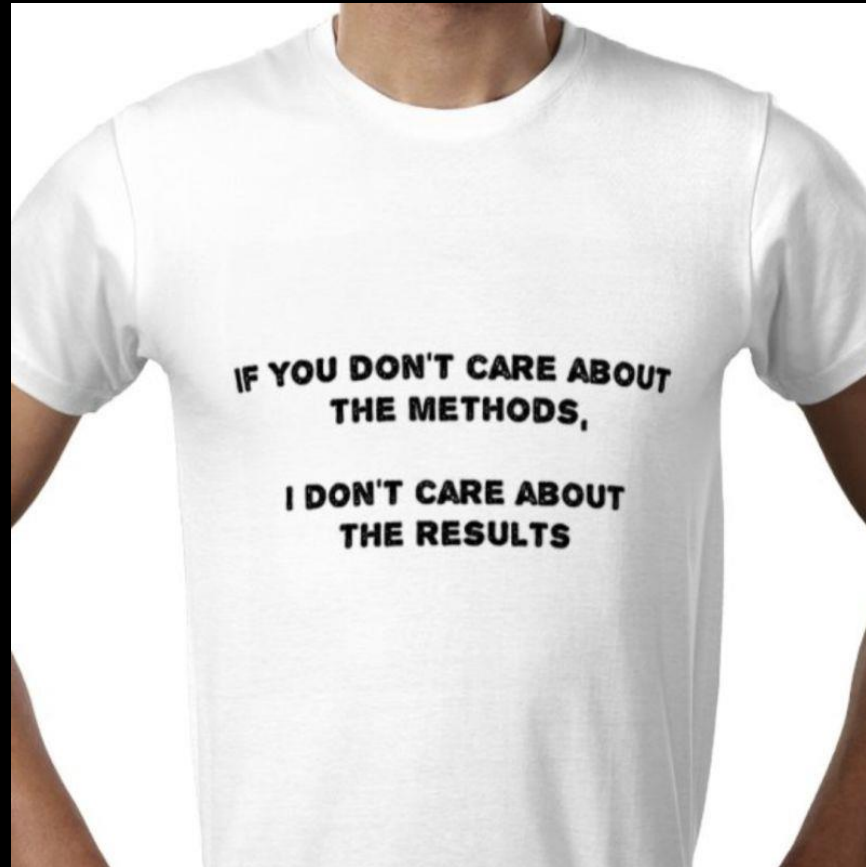
HOW DOES AGE IMPACT MEMORY?

Individual Differences Research:

- Collect chronological age
- Collect memory performance

Quasi-Experimental Research:

- Divide participants by age-group
 - Younger- vs. Older-Adults
- Collect memory performance



Note. From "If you don't care about the methods, I don't care about the results" - the perfect T-shirt for your next scientific conference! [Image attached] [Tweet], by Peter Tennant [@PWGTennant], 2024, June 11, Twitter (twitter.com).



THE USUAL (STATISTICAL) SUSPECTS

ASSESSMENTS OF PREDICTION

Allows for determination of association (i.e., relationship) between 2 variables

Additionally, can predict the value of a variable from another variable or a group of variables



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https://unsplash.com/photos/king-of-diamonds-playing-card-GESOWH4YLRI?utm_source=unsplash&utm_medium=referral&utm_content=creditShareLink

ASSESSMENTS OF PREDICTION

1. Bivariate Correlation

- a) Demonstrates presence of linear relationship (i.e., association) between 2 variables at once
- b) **Required** for next two assessments

2. Simple Linear Regression

- a) Predict 1 variable from another
- b) e.g., Predict memory errors from age

3. Multiple Linear Regression

- a) Predict 1 variable from several variables
- b) e.g., Predict memory errors from age and stress levels



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https://unsplash.com/photos/king-of-diamonds-playing-card-GESOWH4YLRI?utm_source=unsplash&utm_medium=referral&utm_content=creditShareLink

ASSESSMENTS OF PREDICTION

Important:
Correlation DOES NOT
IMPLY Causation

Strongest way to infer causation
is through controlled experimental
research methodology



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https://unsplash.com/photos/king-of-diamonds-playing-card-GESOWH4YLRI?utm_source=unsplash&utm_medium=referral&utm_content=creditShareLink



ASSESSMENTS OF GROUP DIFFERENCES

Allows for assessment of differences due to manipulation or participant grouping(s)

Most methods compare the means of different groups (e.g., experimental conditions) but some consider paired differences



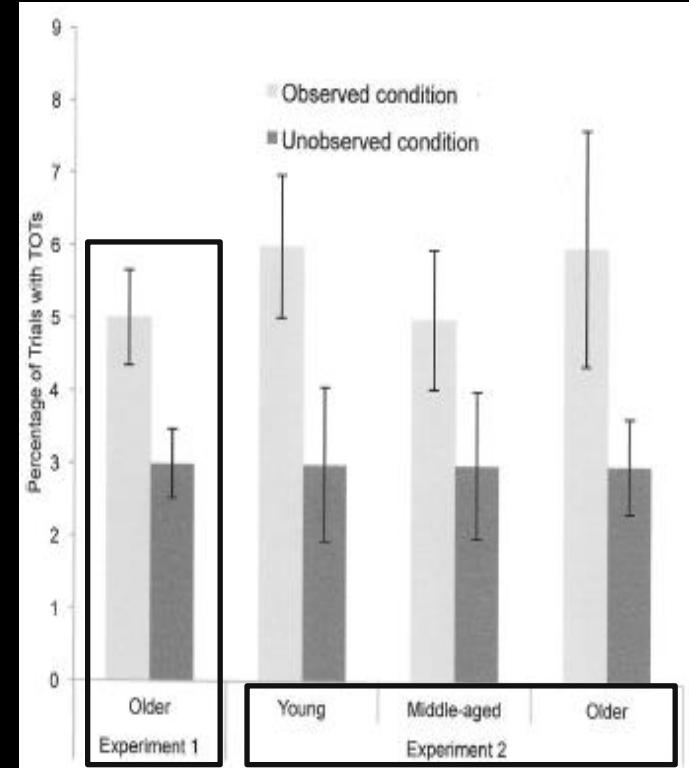
ASSESSMENTS OF GROUP DIFFERENCES

1. Two Groups

- a) Referred as t -tests
- b) Independent versus Dependent (i.e., paired) Samples
- c) Dichotomous IV; Continuous DV

2. Three or More Groups

- a) Referred as One-way ANOVA or F -test
- b) Between- versus Within-Subjects
- c) Categorical IV (3 + levels); Continuous DV

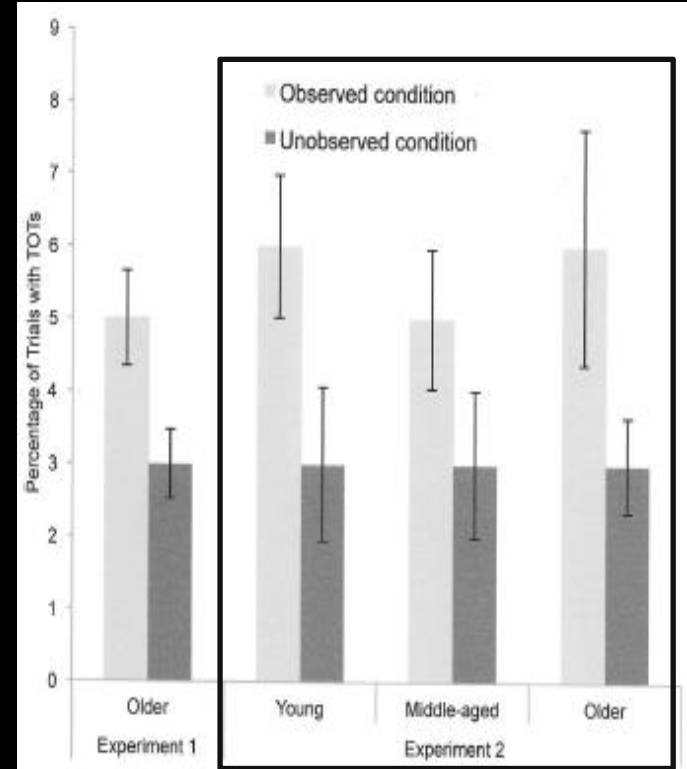


ASSESSMENTS OF GROUP DIFFERENCES

3. Factorial Designs

- a) Referred as Factorial or $\#$ -way[†] ANOVAs
 - a) Requires multiple F -tests
- b) Determines differences in a single DV across multiple IVs simultaneously
 - a) 2 or more categorical IVs (2 + levels)
 - b) Continuous DV

[†] Two-way (2 IVs); Three-way (3 IVs); Four-way (4 IVs) ...





TESTS OF PREDICTION

Appropriate when research question(s) imply Individual Differences or Longitudinal research methodology

TESTS OF GROUP DIFFERENCES

Appropriate when research question(s) imply (Quasi)-Experimental or Longitudinal research methodology

THANK YOU!!

QUESTIONS?

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